

Using experiments to evaluate models of photosynthetic acclimation to global change

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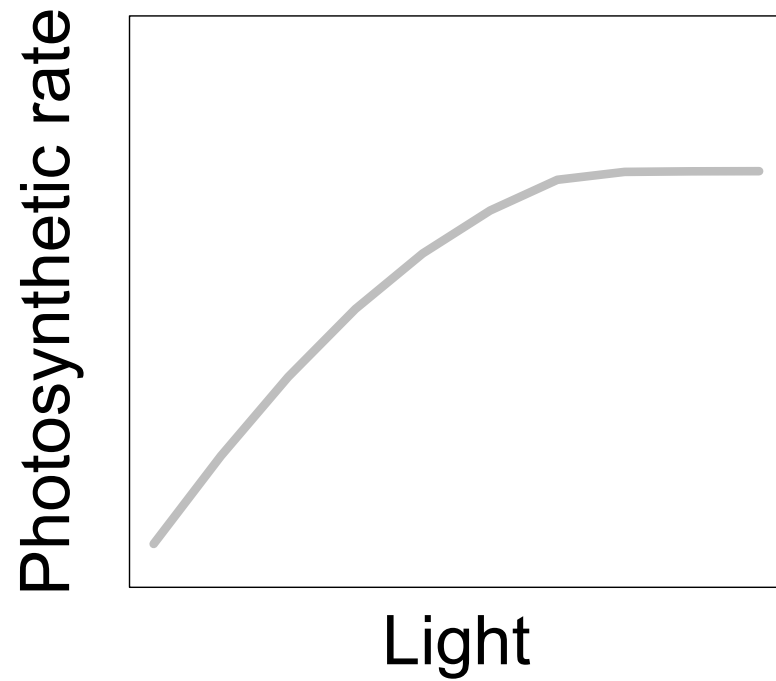


What is photosynthetic
acclimation?

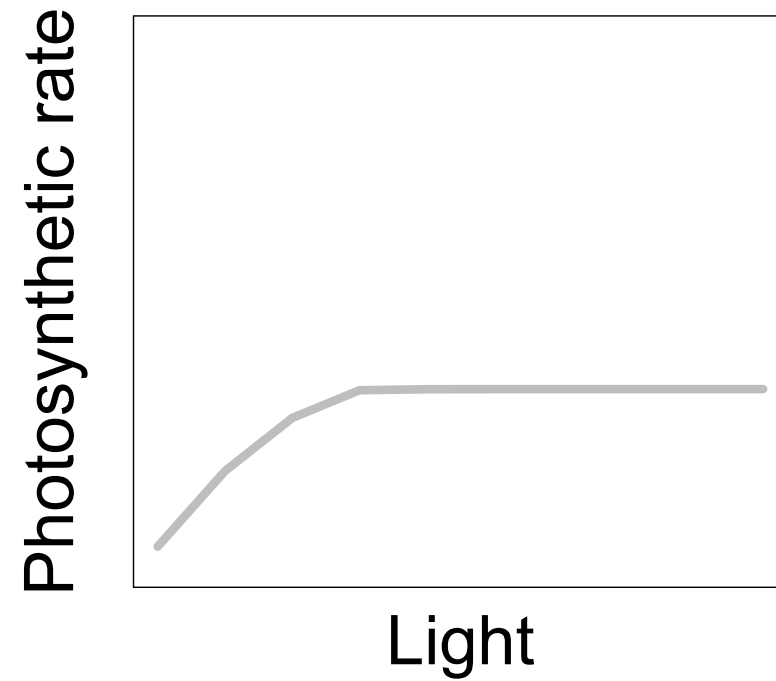
“A physiological, structural, or biochemical adjustment by an individual plant in response to an experimentally induced environmental stimulus that is manifested as alterations in the short-term response function of a physiological process.”

Acclimation: an example

Acclimated to high light

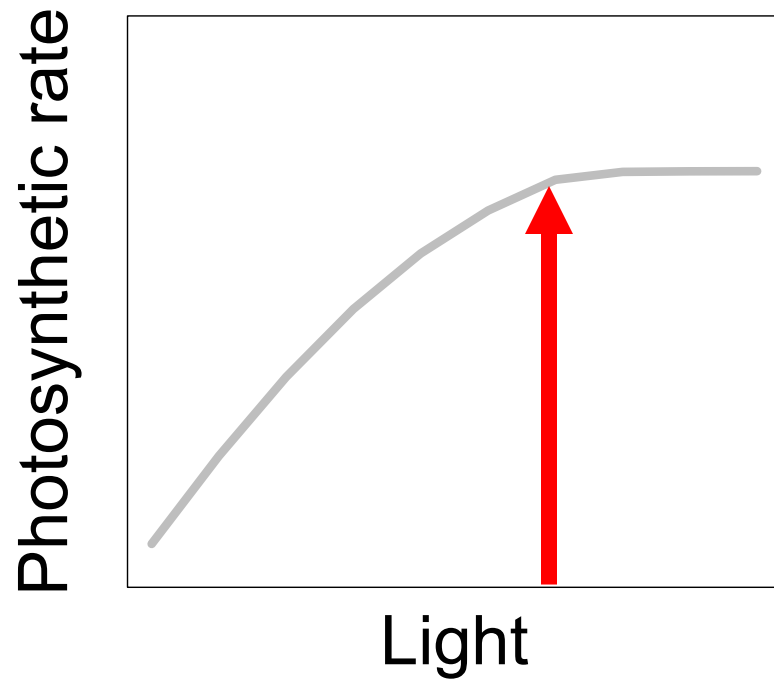


Acclimated to low light

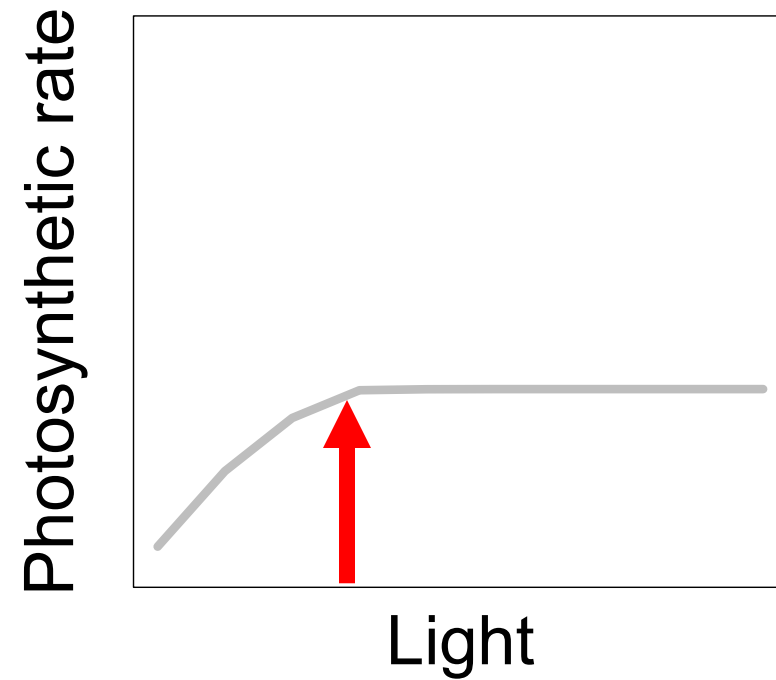


Acclimation: an example

Acclimated to high light



Acclimated to low light



Photosynthesis peaks a light level close to the level the leaf is acclimated to

Acclimation is ubiquitous and well known...

CO₂: Bazzaz (1990)

Annu. Rev. Ecol. Syst. 1990. 21:167–96
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THE RESPONSE OF NATURAL
ECOSYSTEMS TO THE RISING
GLOBAL CO₂ LEVELS

F. A. Bazzaz

Light: Boardman (1977)

Ann. Rev. Plant Physiol. 1977. 28:355–77
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COMPARATIVE
PHOTOSYNTHESIS OF SUN
AND SHADE PLANTS

N. K. Boardman

Division of Plant Industry, CSIRO, Canberra City, A.C.T. 2601, Australia

Temperature: Berry &
Björkman (1980)

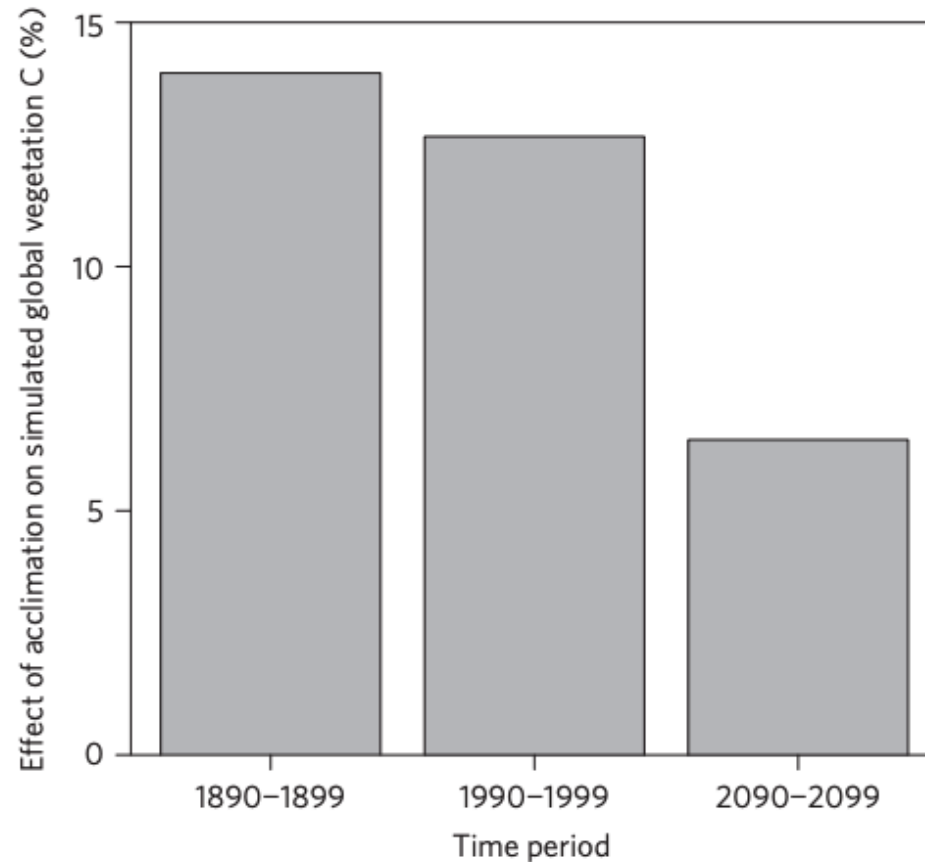
Ann. Rev. Plant Physiol. 1980. 31:491–543
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PHOTOSYNTHETIC RESPONSE
AND ADAPTATION
TO TEMPERATURE
IN HIGHER PLANTS

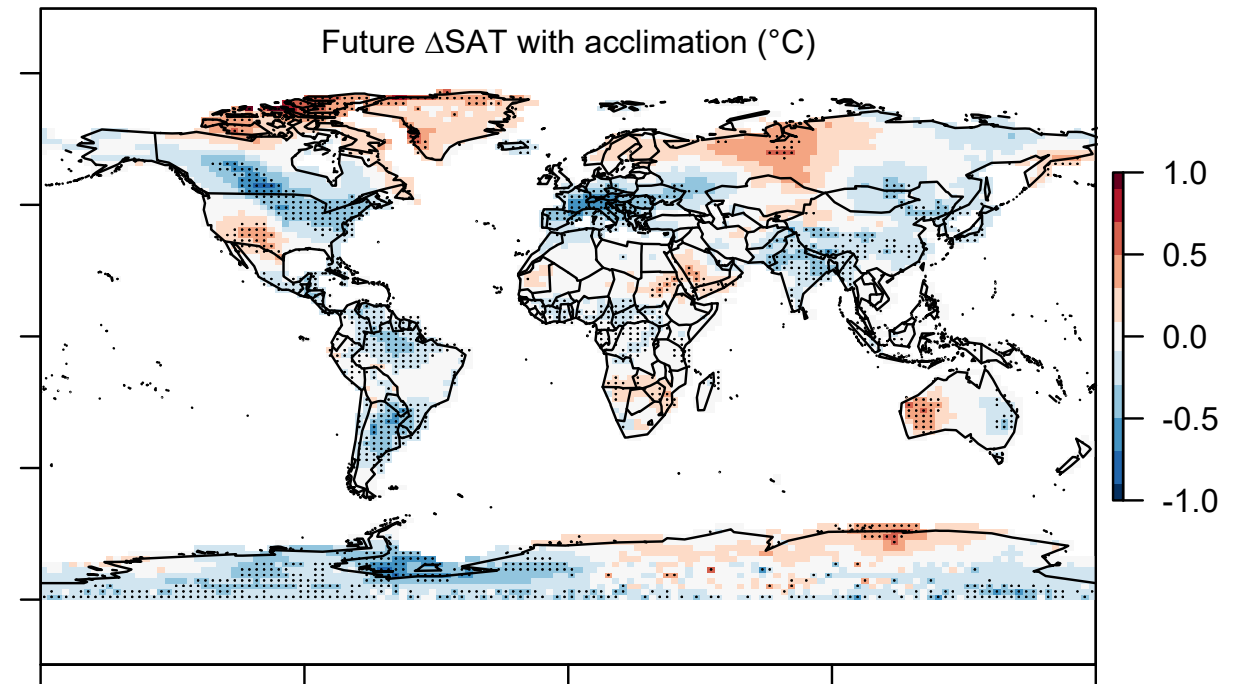
- *Joseph Berry and Olle Björkman*¹

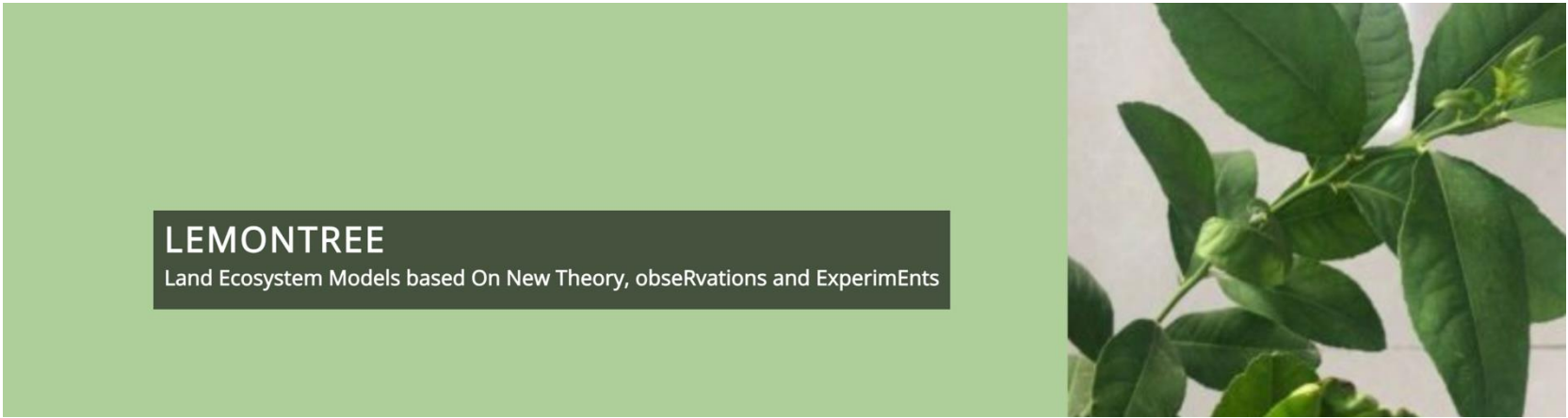
...and can impact carbon cycling and climate

Acclimation increase future C storage by ~6%



Acclimation alters future temperature by $>1^{\circ}\text{C}$



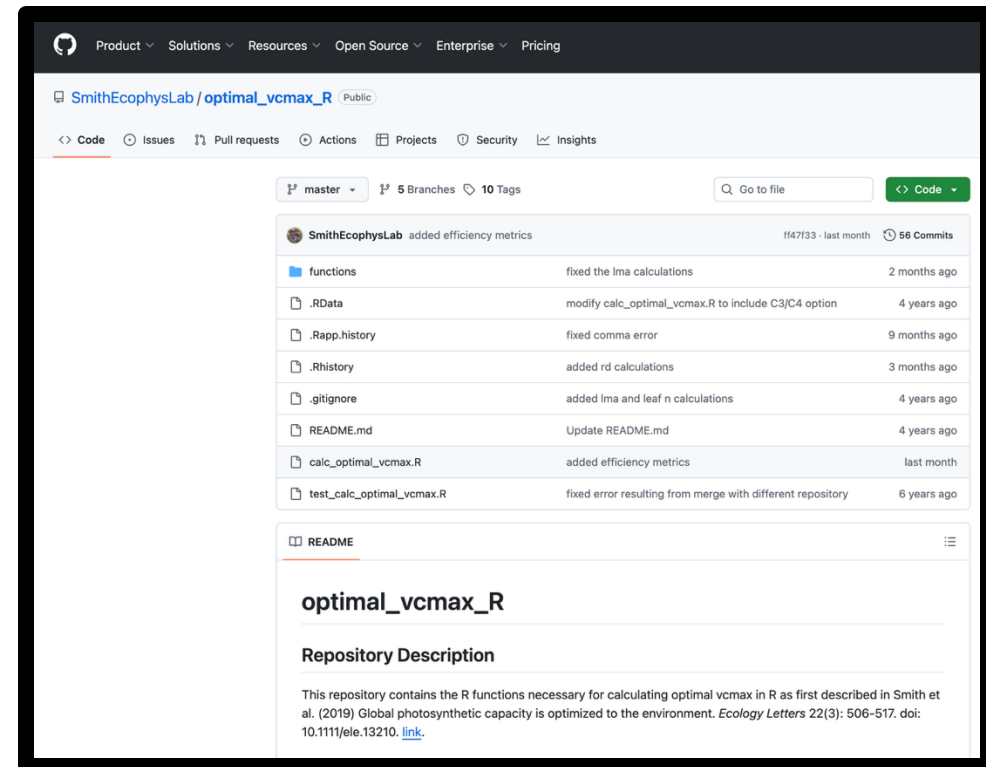


Tansley review | [Free Access](#)

Eco-evolutionary optimality as a means to improve vegetation and land-surface models

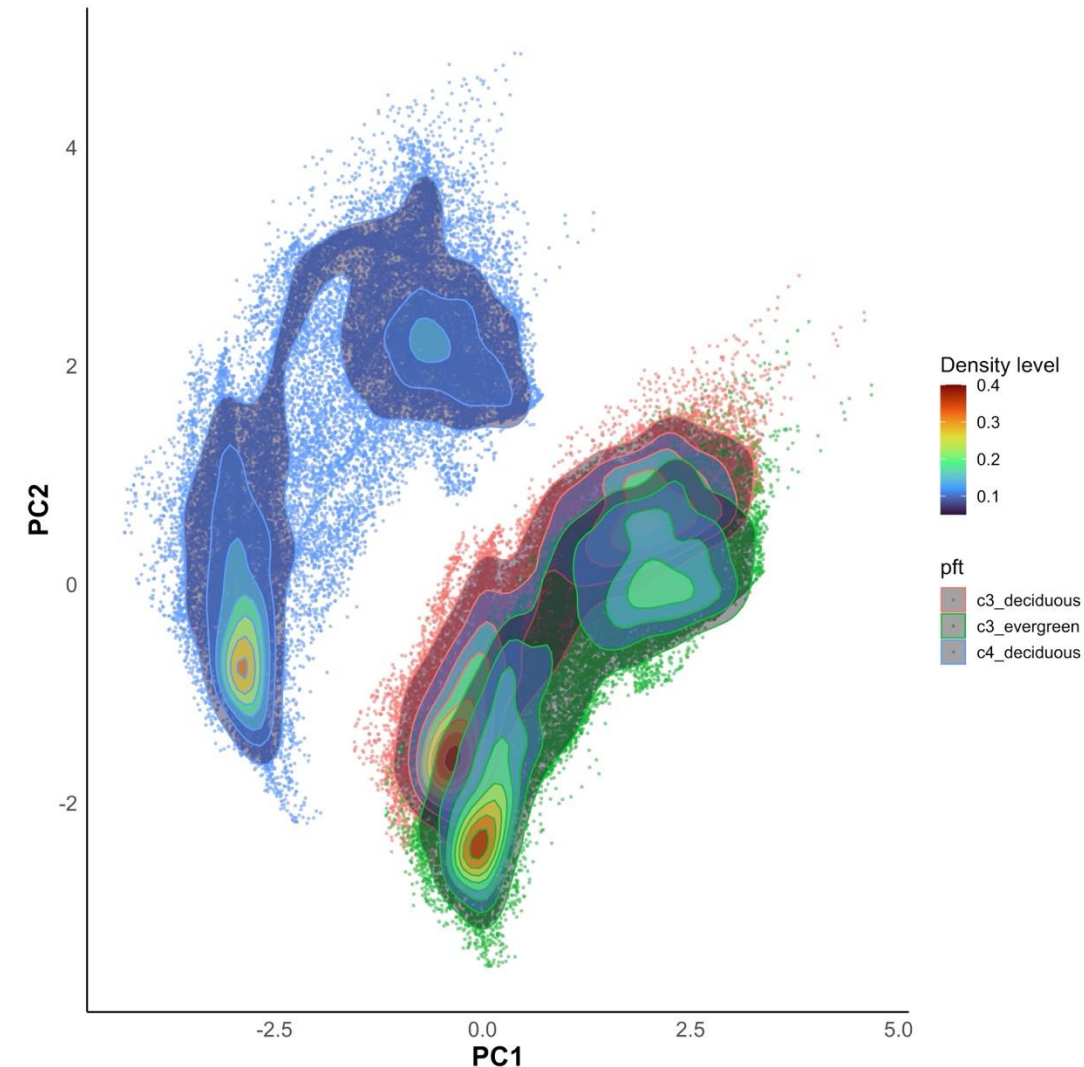
Sandy P. Harrison , Wolfgang Cramer, Oskar Franklin, Iain Colin Prentice, Han Wang, Åke Brännström, Hugo de Boer, Ulf Dieckmann, Jaideep Joshi, Trevor F. Keenan, Aliénor Lavergne, Stefano Manzoni, Giulia Mengoli, Catherine Morfopoulos, Josep Peñuelas, Stephan Pietsch, Karin T. Rebel, Youngryel Ryu, Nicholas G. Smith, Benjamin D. Stocker, Ian J. Wright ... [See fewer authors](#) ^

First published: 15 June 2021 | <https://doi.org/10.1111/nph.17558>



LEMONTREE-developed EEO model of photosynthetic acclimation

- Inputs
 - Environment (T, PAR, VPD, CO₂)
 - Plant type (C3/C4, deciduousness)
- Outputs
 - Photosynthetic capacity
 - Stomatal conductance
 - Leaf mass per area
 - Leaf nitrogen pools
 - Photosynthetic resource use efficiency



How do we know if the model is
producing reliable simulations?

Testing model predictions

1. Compare output to observations

- Pros: data are from "real world"
- Cons: multiple interacting factors at play

Testing model predictions

1. Compare output to observations

- Pros: data are from "real world"
- Cons: multiple interacting factors at play

2. Experimentally alter conditions and compare responses

- Pros: can test factors directly
- Cons: level of "realism" may vary by experiment

Experiment types

- Growth chamber experiments



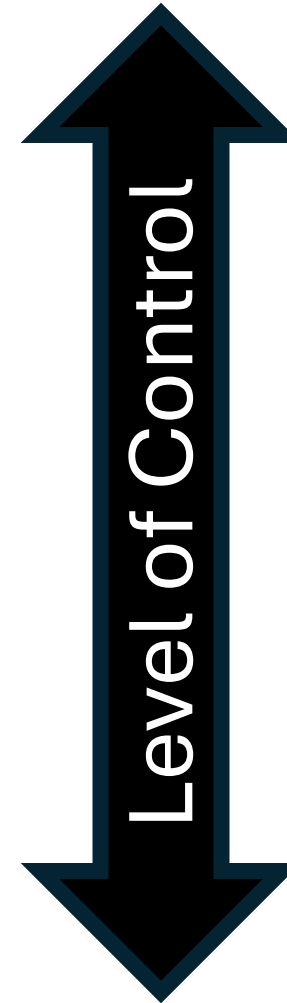
- Glasshouse experiments



- Field manipulation experiments

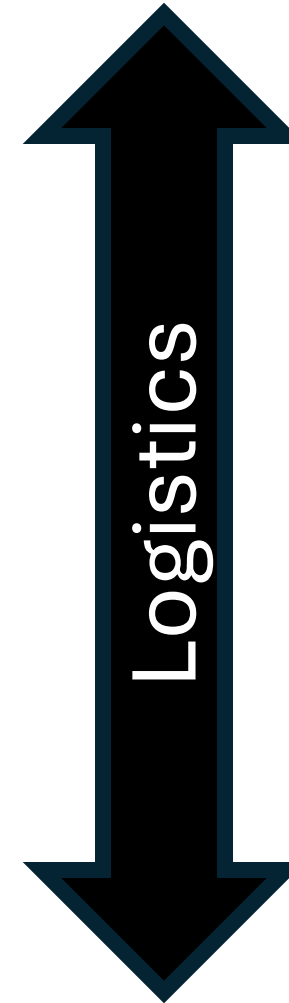


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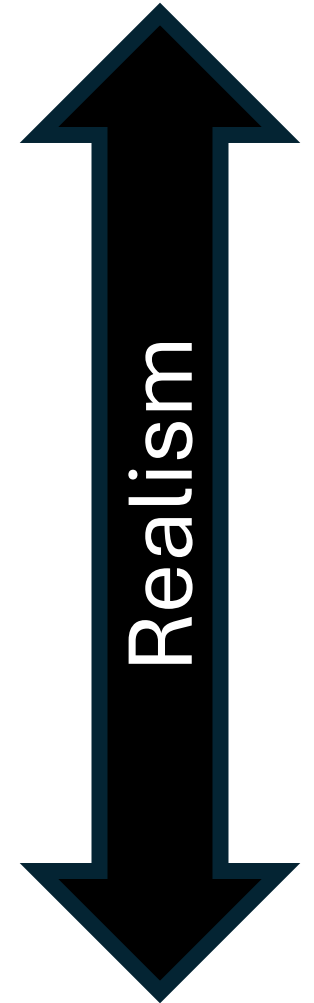
Low

Easy



Hard

Fake



Real

Experiment types

- Growth chamber experiments



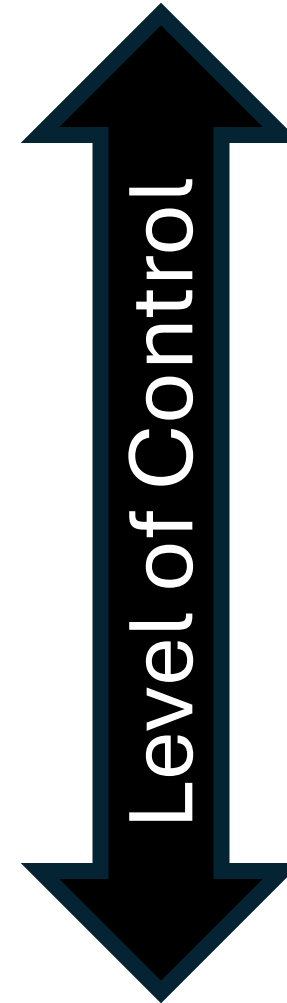
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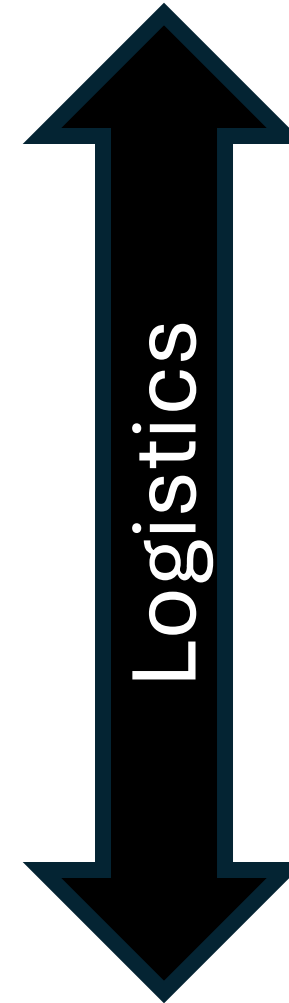


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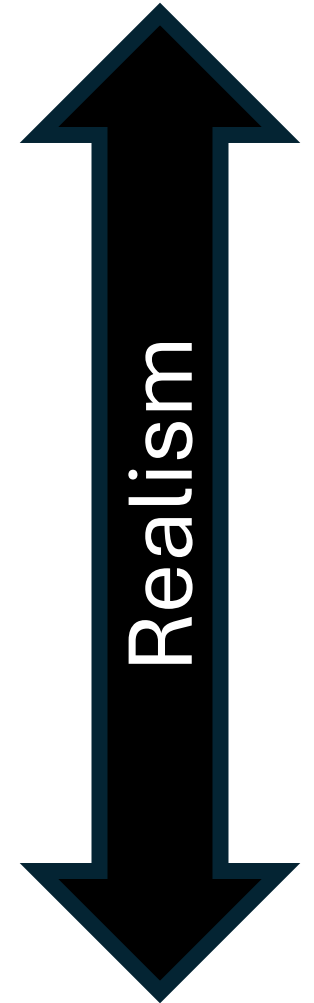
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Testing theory with experiments: **field manipulation** example (Cheaib et al., 2025)

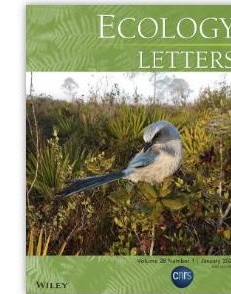


LETTER |  Full Access

Soil Nitrogen Supply Exerts Largest Influence on Leaf Nitrogen in Environments with the Greatest Leaf Nitrogen Demand

Alissar Cheaib , Elizabeth F. Waring, Risa McNellis, Evan A. Perkowski, Jason P. Martina, Eric W. Seabloom, Elizabeth T. Borer, Peter A. Wilfahrt, Ning Dong, Iain Colin Prentice ... [See all authors](#) ▾

First published: 17 January 2025 | <https://doi.org/10.1111/ele.70015> | Citations: 1

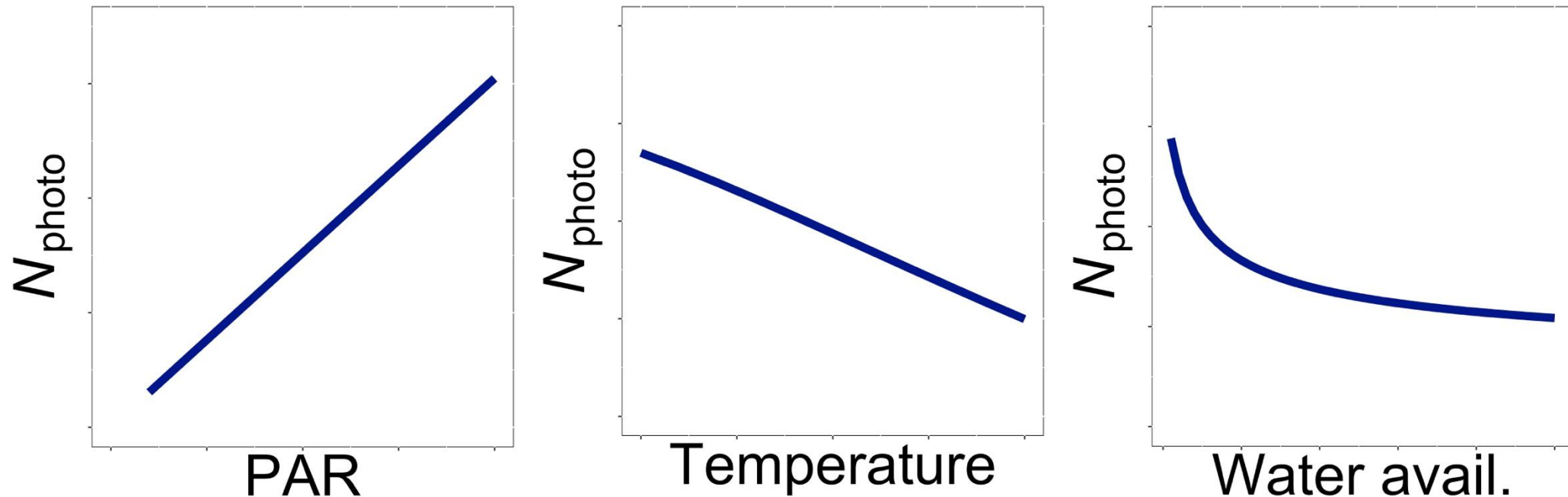


Volume 28, Issue 1
January 2025
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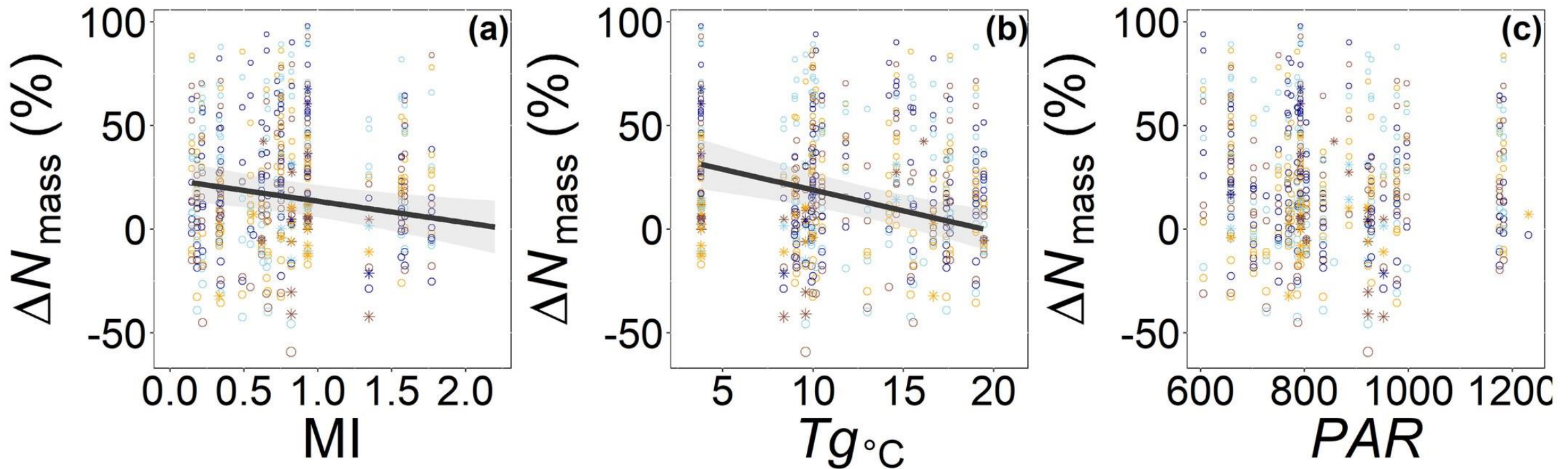
Testing theory with experiments: **field manipulation** example (Cheaib et al., 2025)

- **Theory**: photosynthetic nitrogen demand should increase with light and decrease with temperature and water availability



Testing theory with experiments: **field manipulation** example (Cheaib et al., 2025)

- **Results**: theory confirmed for temperature and water, but not light



Testing theory with experiments: **field manipulation** example (Cheaib et al., 2025)

- Summary
 - Multi-site field manipulation data can be useful for testing theory
 - Such large-scale, field data are difficult to obtain
 - Particularly for “hard” traits
 - Directly testing mechanisms requires complimentary approaches

Ongoing **field manipulation** work

- **Individual and community** trait responses to soil resource availability across climatic gradients (PhD Monika Kelley)
 - Physiological, morphological, and biochemical traits
- Connection between plant trait-environment relationships and **population dynamics** (postdoc Alissar Cheaib)
 - New approach can explore how trait responses influence population dynamics
- Plant trait responses to experimental warming across **elevation** (postdoc Alissar Cheaib)
 - Temperature and non-temperature drivers of elevational responses

Experiment types

- Growth chamber experiments



- Glasshouse experiments



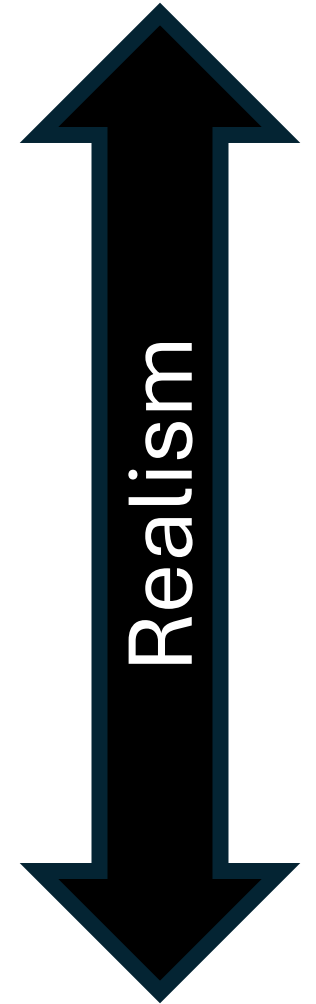
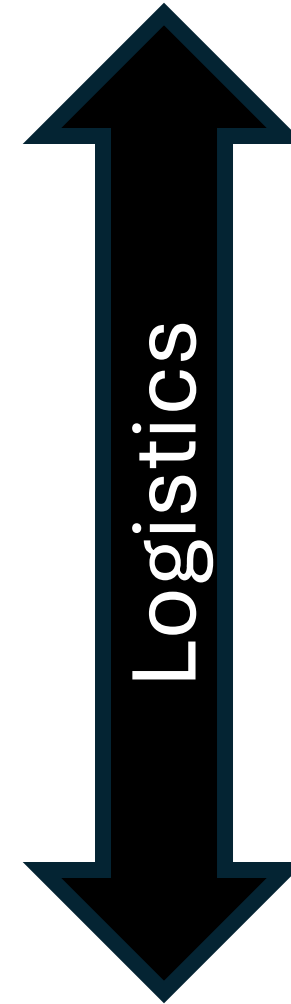
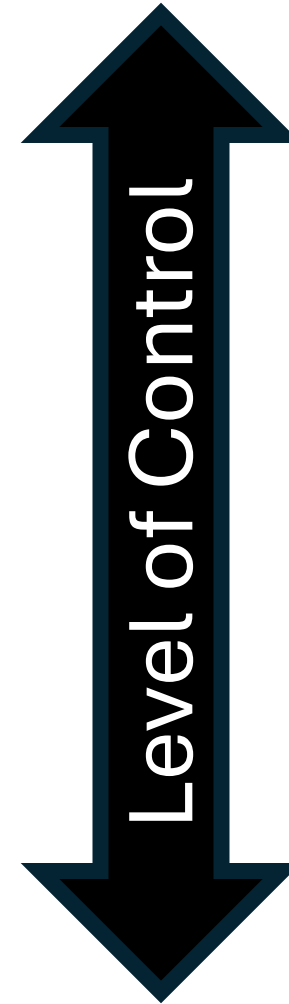
- Field manipulation experiments



High

Easy

Fake



Low

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Testing theory with experiments: glasshouse example (Waring et al., 2023)



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Volume 74, Issue 17
13 September 2023

JOURNAL ARTICLE

Soil nitrogen fertilization reduces relative leaf nitrogen allocation to photosynthesis FREE

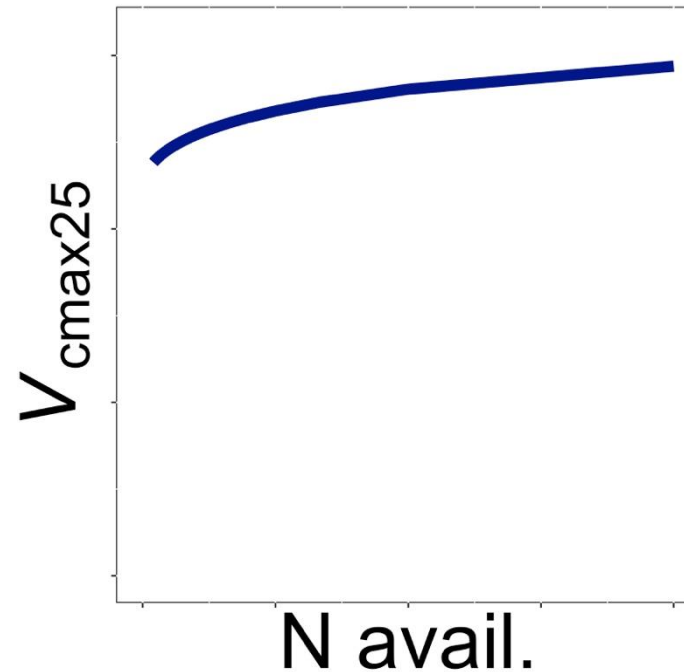
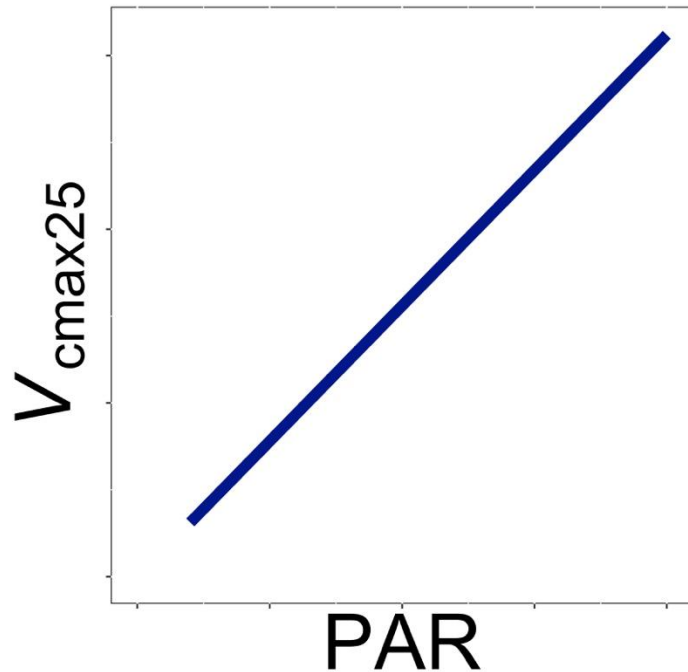
Elizabeth F Waring , Evan A Perkowski , Nicholas G Smith ✉

Journal of Experimental Botany, Volume 74, Issue 17, 13 September 2023, Pages 5166–5180, <https://doi.org/10.1093/jxb/erad195>

Published: 26 May 2023 **Article history** ▼

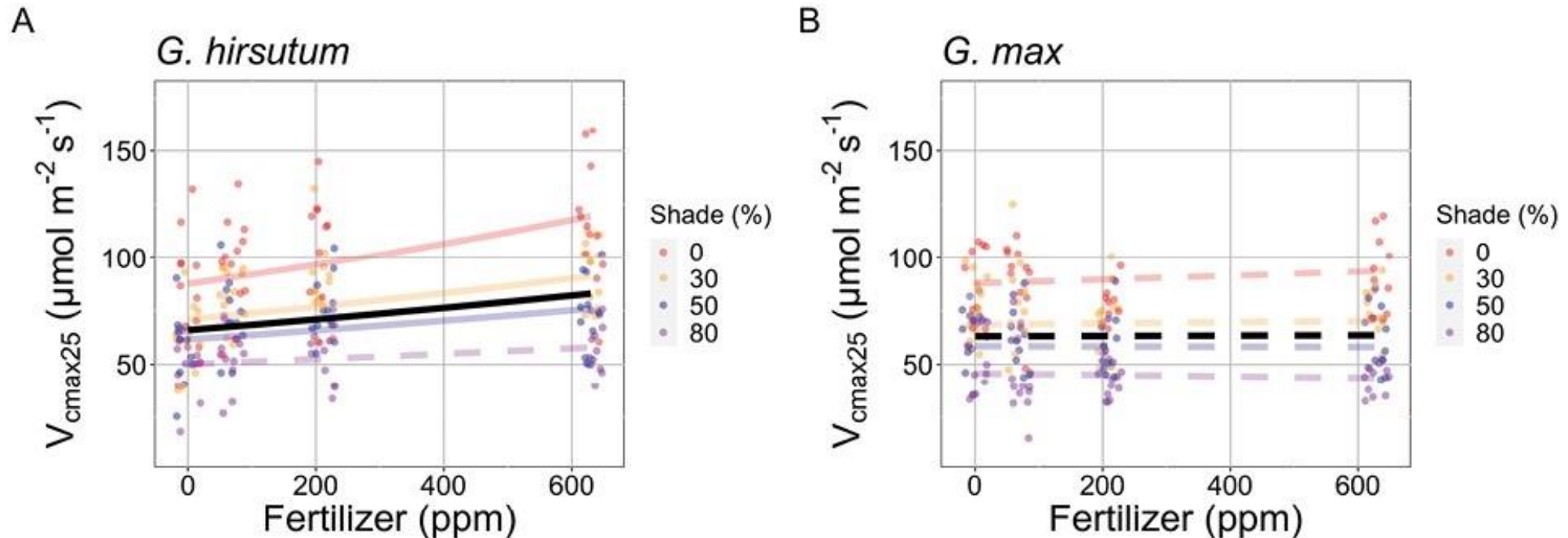
Testing theory with experiments: **glasshouse** example (Waring et al., 2023)

- **Theory**: photosynthetic capacity increases strongly with light, but weakly with soil nitrogen availability



Testing theory with experiments: **glasshouse** example (Waring et al., 2023)

- **Results**: theory confirmed, with additional insight into species differences



Testing theory with experiments: glasshouse example (Waring et al., 2023)

- **Summary**

- Can test direct physiological process responses to altered environmental conditions
- Limitation in conditions that can be considered

Ongoing **glasshouse** work

- Separation of soil **water and nutrients** on plant traits (postdoc Alissar Cheaib)
 - Explores how water influences nutrient availability and impact on theoretical understanding
- Plant trait responses to **phosphorus** demand and availability in C3/C4 species (PhD Monika Kelley)
 - Will be comparable to Waring et al. (2023) results, but for P
- Plant trait responses to environment under **mycorrhizal associations** (PhD Snehanjana Chatterjee and postdoc Alissar Cheaib)
 - Multiple projects using different mycorrhizal and environmental treatments

Experiment types

- Growth chamber experiments



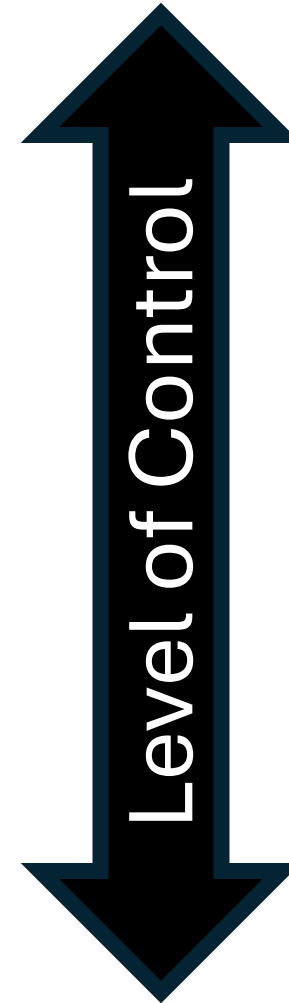
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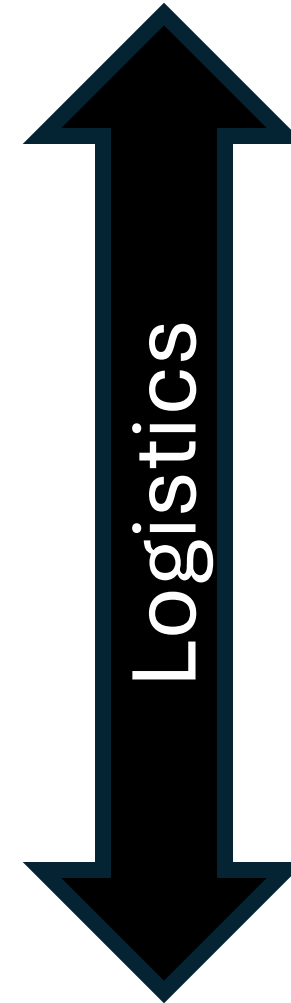


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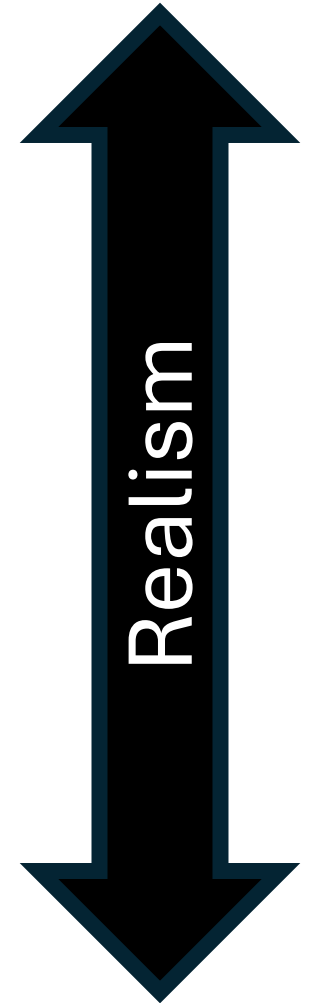
Low

Easy



Hard

Fake



Real

Testing theory with experiments: **growth chamber** example (Perkowski et al., 2025)



Volume 76, Issue 10
2 July 2025

JOURNAL ARTICLE

Nitrogen demand, availability, and acquisition strategy control plant responses to elevated CO₂

Evan A Perkowski ✉, Ezinwanne Ezekannagha, Nicholas G Smith

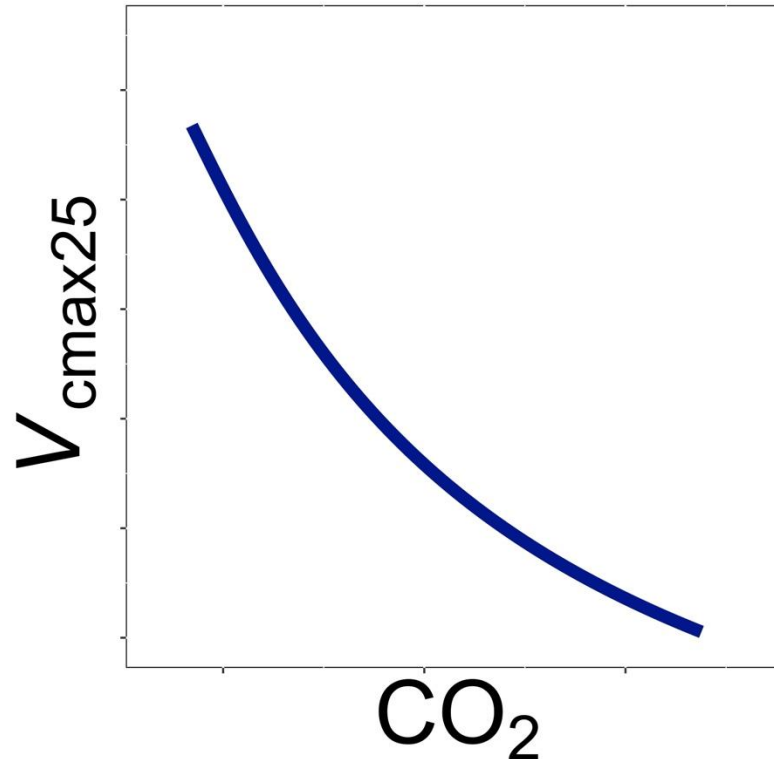
Journal of Experimental Botany, Volume 76, Issue 10, 2 July 2025, Pages 2908–2923,

<https://doi.org/10.1093/jxb/eraf118>

Published: 17 March 2025 **Article history** ▼

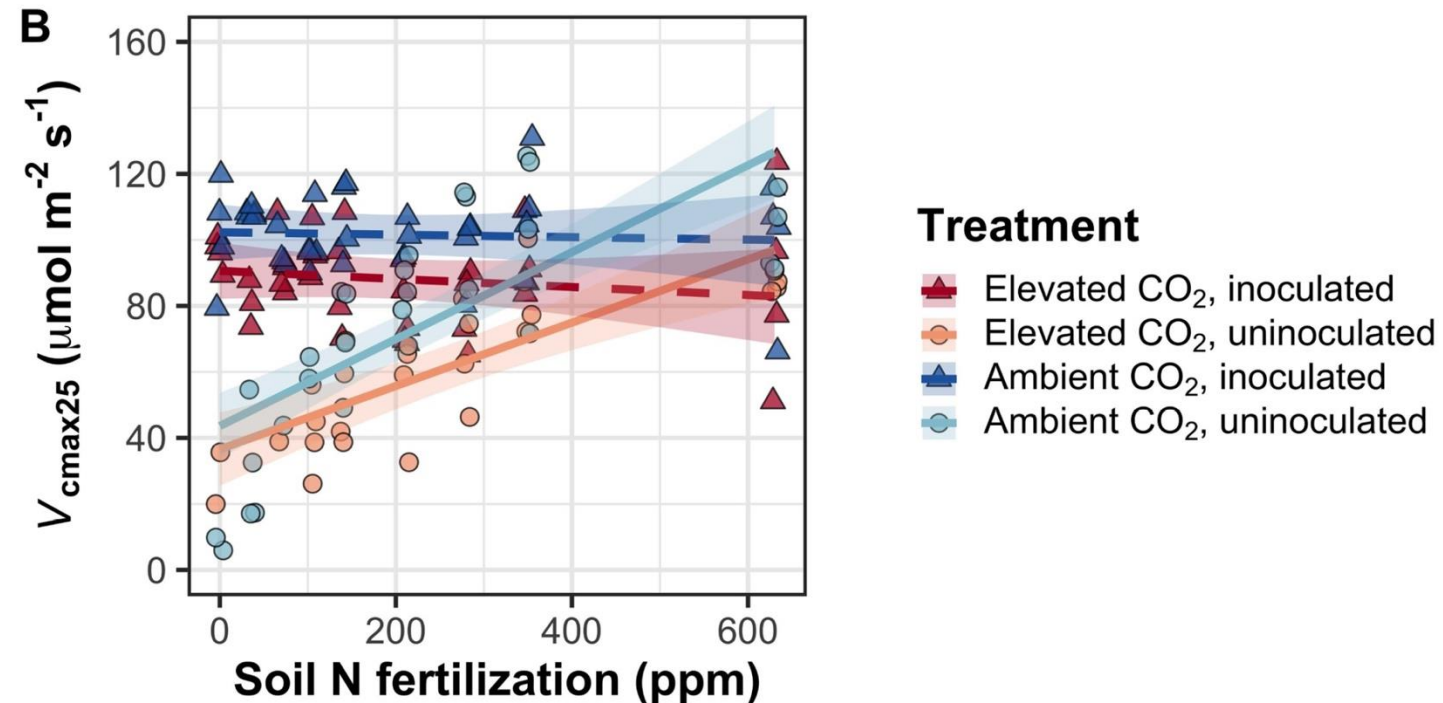
Testing theory with experiments: growth chamber example (Perkowski et al., 2023)

- **Theory:** elevated CO_2 should reduce photosynthetic capacity regardless of context



Testing theory with experiments: **growth chamber** example (Perkowski et al., 2023)

- **Results**: lack of context dependency of CO₂ acclimation confirmed



Testing theory with experiments: growth chamber example (Perkowski et al., 2023)

- Summary

- Growth chambers allow for theory testing under many different environmental conditions
- Realism may need to be checked alongside field data

Ongoing **growth chamber** work

- Testing mechanisms driving **temperature acclimation** responses, including interactions with nutrients (PhD Dan Owusu Kwakye)
 - Hope to explain variable responses seen in literature
- Testing more mechanistic representations of **light reactions** and their acclimation responses (PhD Garry Garza)
 - Including quantum yield and cytochrome b6f
- Comparing temperature and CO₂ acclimation between **C₃ and C₄** grasses (MS Zinny Ezekannagha)
 - Both acclimate to temperature, but C₄ shows little response to CO₂
- Exploring **timescales** and responses to **variability** (MS Eve Gray and Christine Vanginault)

Takehomes

- Multiple approaches are needed to test theory
- Experiments can complement observational data
 - Allows for control of different factors
- Experiments have tradeoffs
 - Level of control
 - Logistics
 - Realism
- Multiple types of experiments may be needed to robustly test mechanisms

Presentation available at:

www.github.com/SmithEcophysLab/seminar/2025_lemontree

Data and code:

www.github.com/SmithEcophysLab

Contact: nick.smith@ttu.edu



Thanks!